Knowing the geometric effects of multiplying complex numbers in different ways makes it much easier to perform otherwise tricky computations.

Suppose we start with some complex number w. Here's a summary of the geometric effects we see when multiplying w by some different values.

Multiply w by...	Geometric effect on w
a	Dilate by a scale factor of $\lvert a \rvert$
i	Rotate 90° counterclockwise about the origin
$a+bi$	Dilate by a scale factor equal to the modulus $\lvert a+bi \rvert$ and rotate counterclockwise about the origin by an angle equal to the argument of $a+bi$

So if we find the modulus and argument of $-\sqrt{3}+i$, we can figure out which point in the graph represents the product $z(-\sqrt{3}+i)$.

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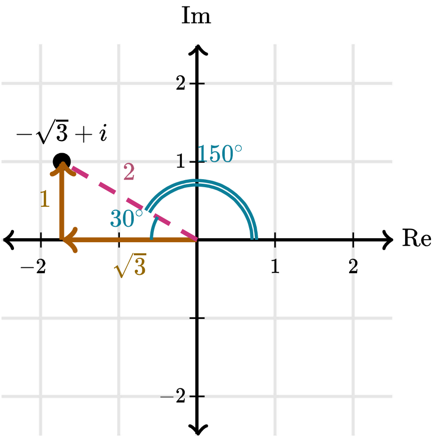
Modulus and argument of $-\sqrt{3}+i$



The modulus of $-\sqrt{3}+i$ is $\sqrt{(\sqrt{3})^2+1^2}=2$.

Since $-\sqrt{3}+i$ creates a special right triangle, we know that its argument must be 150°.

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Geometric effect of multiplication

We can find the general location of product $z(-\sqrt{3}+i)$ by starting at z and rotating 150° counterclockwise about the origin.

Then, we dilate by a scale factor of 2. In other words, the modulus of $z(-\sqrt{3}+i)$ is two times as large as the modulus of z.